

Henrique Ennes

Curriculum Vitae

Personal information

Full name Henrique Lovisi Ennes
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GitHub [HLovisiEnnes](#)
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Education

2023 – **Ph.D. candidate in Computer Science**, *Inria d'Université Côte d'Azur*, Sophia Antipolis, France.
Project title: *Quantum computing in topology*.
Supervisors: Clément Maria and Nicolas Nisse.

2021–2023 **Master's degree in Applied Mathematics**, *Fundação Getulio Vargas - EMAP*, Rio de Janeiro, Brazil., GPA: 4.00/4.00.
Thesis title: *Detection of representation orbits of compact Lie groups on point clouds*.
Supervisors: Raphaël Tinarrage and César Camacho.

2016–2020 **Bachelor of Arts, Mathematics and Physics (summa cum laude)**, *Whitman College*, Walla Walla (WA), USA., GPA: 3.97/4.00.
Honors in majors.

Grants and awards

Master's **CAPES graduate student grant**
Scholarship offered by the Ministry of Education awarded to graduate students in Brazil based on merit criteria.

Undergraduate **Elected member of Phi Beta Kappa honor society**
Oldest academic honor society in the United States.

Laura and John Hook Family Mathematics Award
Award offered to a graduating senior in Mathematics for demonstrated talent in this field of research.

Student Commencement Marshal
Chosen among the 10 highest general GPAs of the junior class (GPA: 3.97/4.00).

M&G Wier Scholarship
Grant awarded to a junior student in recognition of talent in Mathematics.

Tristram S. Lundquist Scholarship Endowment
Grant awarded to a Physics major student in recognition of academic performance in the physical sciences.

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- Others **Mayor's medal of honor for advancing the study of Astronomy in Juiz de Fora**
Award offered for high performance in the Brazilian Astronomy Olympics.

Preprints

- 2025 **Compressed data structures for Heegaard splittings**, with Clément Maria.
Preprint. <https://arxiv.org/abs/2507.11406>.

Publications and papers in conferences

- 2025 **Hardness of computation of quantum invariants on 3-manifolds with restricted topology**, with Clément Maria.
Accepted to the 33rd *European Symposium of Algorithms*. <https://arxiv.org/abs/2503.02814>.
- 2025 **LieDetect: Detection of representation orbits of compact Lie groups from point clouds**, with Raphaël Tinarrage.
Foundations of Computational Mathematics. <https://link.springer.com/article/10.1007/s10208-025-09728-4>
- 2025 **Empirical analysis of Biding Precedent efficiency in the Brazilian Supreme Court via Similar Case Retrieval**, with Raphaël Tinarrage, Lucas E. Resck, Lucas T. Gomes, Jean R. Ponciano, and Jorge Poco.
Artificial Intelligence and Law. <https://doi.org/10.1007/s10506-025-09458-6>.
- 2024 **Cost Benefit Analysis for Investments in Power Grid Resilience - A Guide**, with Edson Daniel Lopes Gonçalves, Joisa Dutra, Rafael Souza.
9th Latin American Energy Economics Meeting (ELAEE)
- 2023 **Measuring the Power Grid Resilience: A Case Study Applied to Brazilian Distribution Companies**, with Joisa Dutra, Rafael Souza, Rafael Gomes, Lucas Amaro, Camila Albertin
27th International Conference on Electricity Distribution.
- 2021 **Two-body bound states through Yukawa forces and perspectives on hydrogen and deuterium**, with Moira Gresham and Alexander Shaw.
American Journal of Physics. <https://doi.org/10.1119/10.0002998>.

Research experience

- 2021–2023 **Researcher in statistics applied to electric power distribution**, *Fundação Getúlio Vargas – Center for Regulatory and Infrastructure Studies*, Rio de Janeiro, Brazil
- 2022–2023 **Researcher in empirical International Law**, *Fundação Getúlio Vargas – ERAS-MUS+ Jean Monnet Centre of Excellence*, Rio de Janeiro, Brazil
- 2021–2023 **Researcher in empirical Constitutional Law**, *Fundação Getúlio Vargas – School of Applied Mathematics*, Rio de Janeiro, Brazil
- 2019–2020 **Researcher in geometric quantization**, *Whitman College*, Walla Walla (WA)
- 2018–2019 **Research assistant in nuclear physics and cosmology**, *Whitman College*, Walla Walla (WA)

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■ Teaching experience

September 2022 **Introduction to Mathematical Modelling Applied to Law**, *Fundação Getúlio Vargas – ERASMUS+ Jean Monnet Centre of Excellence*, Rio de Janeiro, Brazil.
December 2022 (16 hours)

■ Other relevant experiences

August 2020 **Teaching Assistant: Physics 347 (Classical Mechanics)**, *Whitman College*,
December 2020 Walla Walla, WA (remote)
August 2017 **Tutor: Physics 155 and 156 (Introductory Physics Courses)**, *Whitman College*,
May 2020 Walla Walla, WA
January 2018 **Tutor: Mathematics 125, 126, and 225 (Calculus)**, *Whitman College*, Walla
May 2020 Walla, WA

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